

★ WEEK 12 · PART 1 of 3

Interest-Rate Models: The Short Rate

Vasicek and CIR — Last Week's OU Machinery, Applied to Rates

What makes rates different · Vasicek · negative rates · CIR

Rates revert, they don't wander. We apply the Ornstein-Uhlenbeck process to the most important quantity in fixed income — the short rate — and meet the two classic models built on it.

1

PART 1 · THE SHORT RATE

Vasicek, CIR, and Mean-Reverting Rates

What sets interest rates apart, the short-rate modeling idea, the Vasicek model as OU for the rate, its negative-rate catch, and the CIR square-root diffusion that fixes it.

When Rates Went Wild — Part 1

Why a stock model can't price a bond

You cannot price a bond with a stock model. A stock can drift up forever; an interest rate cannot. Rates are pulled back toward a long-run level, and every bond's value depends on the whole future path of that rate.

Why stocks and rates differ

- A stock price can trend away without limit.
- A rate is tethered — it reverts to a normal level.
- GBM captures the first; it cannot capture the second.

Why it matters for bonds

- A bond's price is the discounted future short rate.
- So you must model how the rate EVOLVES, not just today's.
- That needs a mean-reverting model of the rate.

When Rates Went Wild — Part 2

Vasicek, CIR, and the lesson

In 1977 Oldrich Vasicek modeled the short rate as an OU process — mean-reverting, but able to go negative. In 1985 Cox, Ingersoll and Ross added a square-root diffusion to keep it positive. The lesson: the diffusion matters as much as the drift.

The two models

- Vasicek: OU applied directly to the rate.
- CIR: the same drift, a square-root diffusion.
- Both are mean-reverting; they differ in the noise term.

The lesson

- Choosing the drift is only half the model.
- The diffusion decides whether rates can go negative.
- Small SDE choices have large financial consequences.

From OU to Interest Rates

Where Week 11 leads

1

We built the OU process

A mean-reverting SDE: a spring pulling a state back to a long-run mean, with three parameters and a half-life.

2

Interest rates are the natural application

The short rate reverts to a normal level rather than wandering. It is exactly what OU was built to describe.

3

Two classic short-rate models follow

Vasicek is OU for the rate; CIR modifies the diffusion so rates stay positive. This week turns OU into fixed-income machinery.

Part 1 in one line: apply the OU process to the short rate to get Vasicek and CIR.

What Makes Interest Rates Different

Not just another asset price

Interest rates behave unlike stock prices. They revert to a normal level rather than trending; they can't grow without bound; and a single rate today implies a whole curve of yields across maturities. Modeling them needs its own toolkit.

How rates behave

- They revert toward a long-run, policy-anchored level.
- They don't compound upward like a stock.
- They move within a bounded, economically-set range.

What that implies

- A mean-reverting model, not GBM.
- One rate must generate yields for all maturities.
- The whole future path of the rate matters.

The Short-Rate Modeling Idea

One rate to generate them all

The setup

- Model a single instantaneous 'short rate'.
- Everything else is derived from its dynamics.
- Bond prices come from discounting at this rate.

Why it is powerful

- One SDE implies an entire yield curve.
- A few parameters describe the whole term structure.
- It is the foundation of classical rate modeling.

The Vasicek Model

The OU process, applied to the short rate

$$\text{Vasicek: } dr_t = \kappa(\theta - r_t) dt + \sigma dW_t$$

What it says

- The rate is pulled toward a long-run level θ .
- The pull has strength κ — the reversion speed.
- A constant-size Brownian shock adds randomness.

Why it works

- It is literally the OU process, now for the rate.
- Mean reversion matches how rates actually behave.
- It yields closed-form bond prices — very tractable.

Vasicek Is OU for the Short Rate

Nothing new — just a new application

The mapping

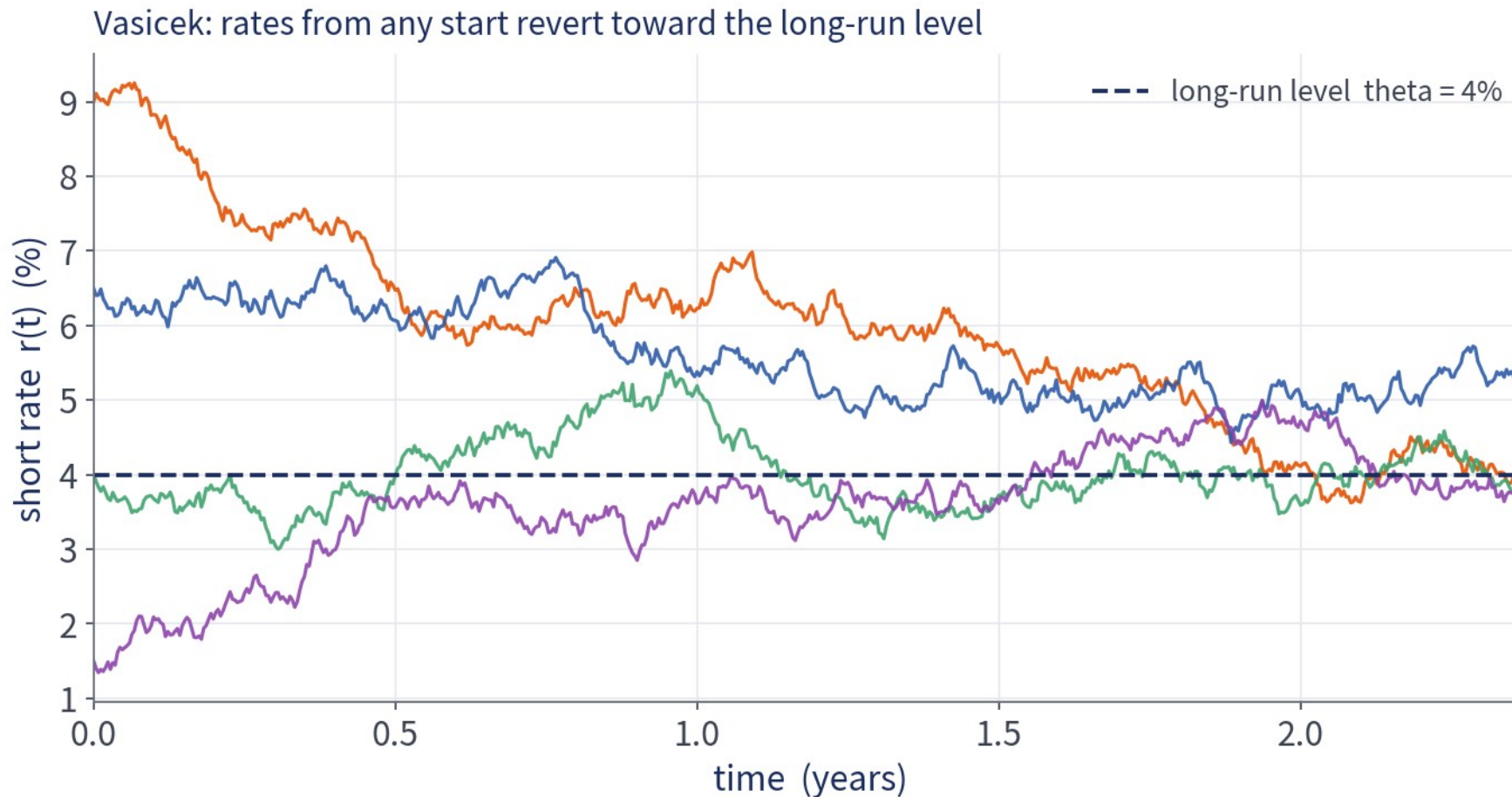
- The state X becomes the short rate r .
- θ is the long-run rate; κ the reversion speed.
- σ is the rate's volatility.

What carries over

- The half-life, $\log_2$ over κ , still applies.
- The stationary distribution is still Gaussian.
- Estimation still runs through the AR(1) bridge.

Vasicek = OU for the Short Rate

Rates starting high, low, or on target are all pulled toward the long-run level θ . The same mean-reverting spring from Week 11, now describing the most important rate in finance.



Vasicek's Mean Reversion

Rates pulled home to a long-run level

The long-run level

- Theta is the level rates revert toward.
- It reflects the normal, policy-anchored rate.
- Deviations are temporary, not permanent.

The speed of return

- Kappa sets how fast rates come back.
- The half-life is $\log_2$ over kappa.
- Faster reversion means shorter-lived swings.

Vasicek's Catch: Negative Rates

Constant volatility can push rates below zero

Vasicek's shock has a constant size regardless of the rate's level. So even when the rate is near zero, a big enough downward shock can push it below zero — something long seen as a flaw, though real rates did turn negative in the 2010s.

Why it happens

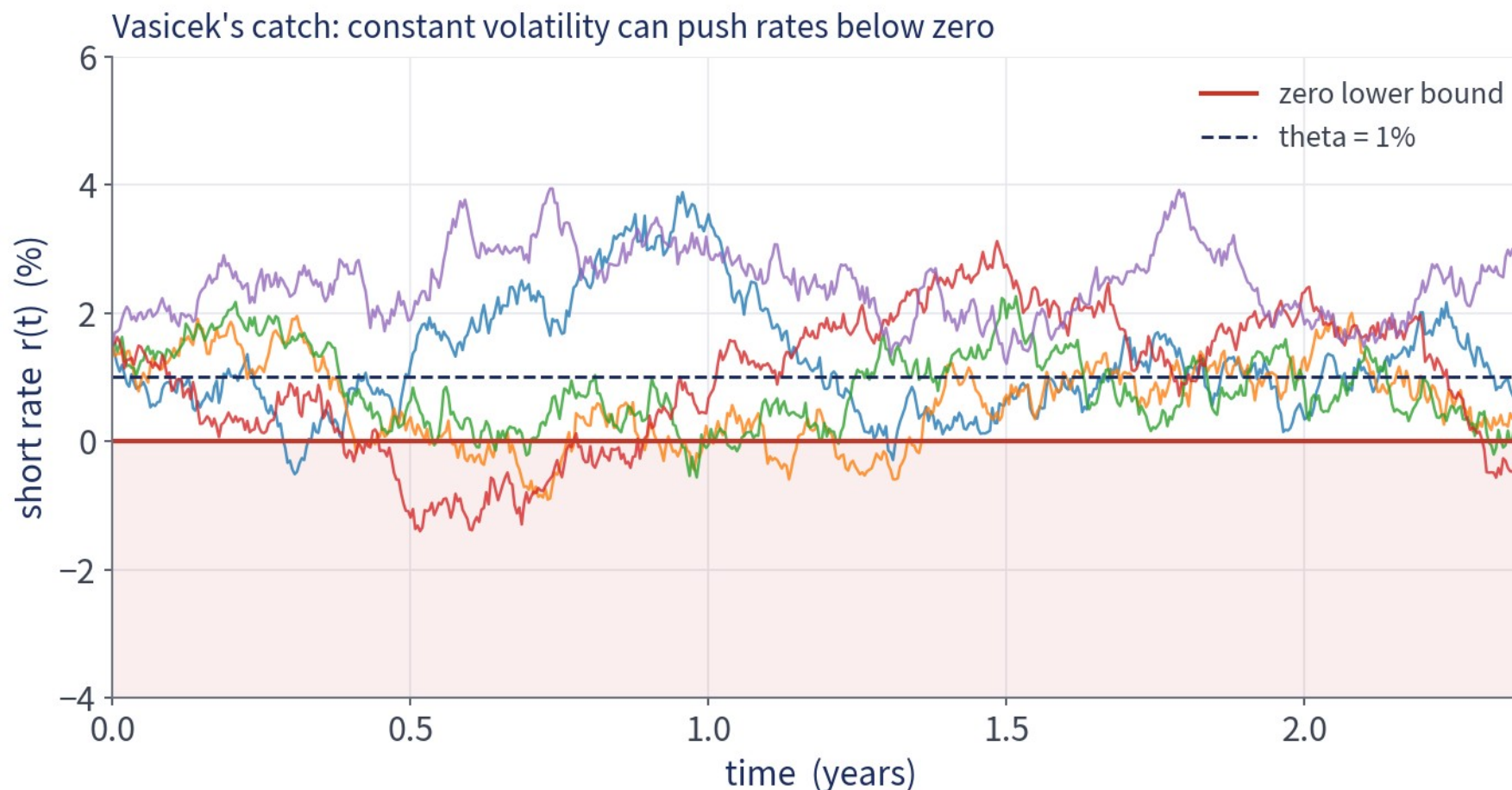
- The diffusion is constant — it doesn't shrink near zero.
- A large negative shock isn't damped at low rates.
- The Gaussian distribution has support below zero.

How much it matters

- For decades it was treated as an obvious defect.
- But some economies did see negative policy rates.
- Still, many applications want a strictly positive rate.

The Negative-Rate Problem, Visualized

With a low long-run level and constant volatility, several Vasicek paths dip below zero (shaded). The constant diffusion offers no floor — the model has no built-in respect for a zero bound.



The CIR Model

A square-root diffusion keeps rates positive

$$\text{CIR: } dr_t = \kappa(\theta - r_t) dt + \sigma \sqrt{r_t} dW_t$$

What changed

- The drift is identical to Vasicek — same mean reversion.
- The diffusion is now sigma times the square root of the rate.
- So volatility shrinks as the rate approaches zero.

The effect

- Near zero, shocks become tiny and can't push through.
- The rate is repelled from the zero boundary.
- It stays non-negative, under a mild condition.

The Square-Root Diffusion

Why the noise term is the whole story

diffusion:

$\underbrace{\sigma}_{\text{Vasicek: constant}}$

VS

$\underbrace{\sigma\sqrt{r_t}}_{\text{CIR: shrinks as } r \rightarrow 0}$

The contrast

- Vasicek: a constant diffusion at every rate level.
- CIR: a diffusion that scales with the root of the rate.
- Same drift, entirely different behavior near zero.

Why it helps

- As the rate falls to zero, its volatility vanishes.
- So the rate cannot be knocked negative.
- Volatility also rises with the level, matching data.

Why CIR Stays Positive

The Feller condition

Feller condition: $2\kappa\theta \geq \sigma^2 \Rightarrow r_t > 0 \text{ always}$

What it says

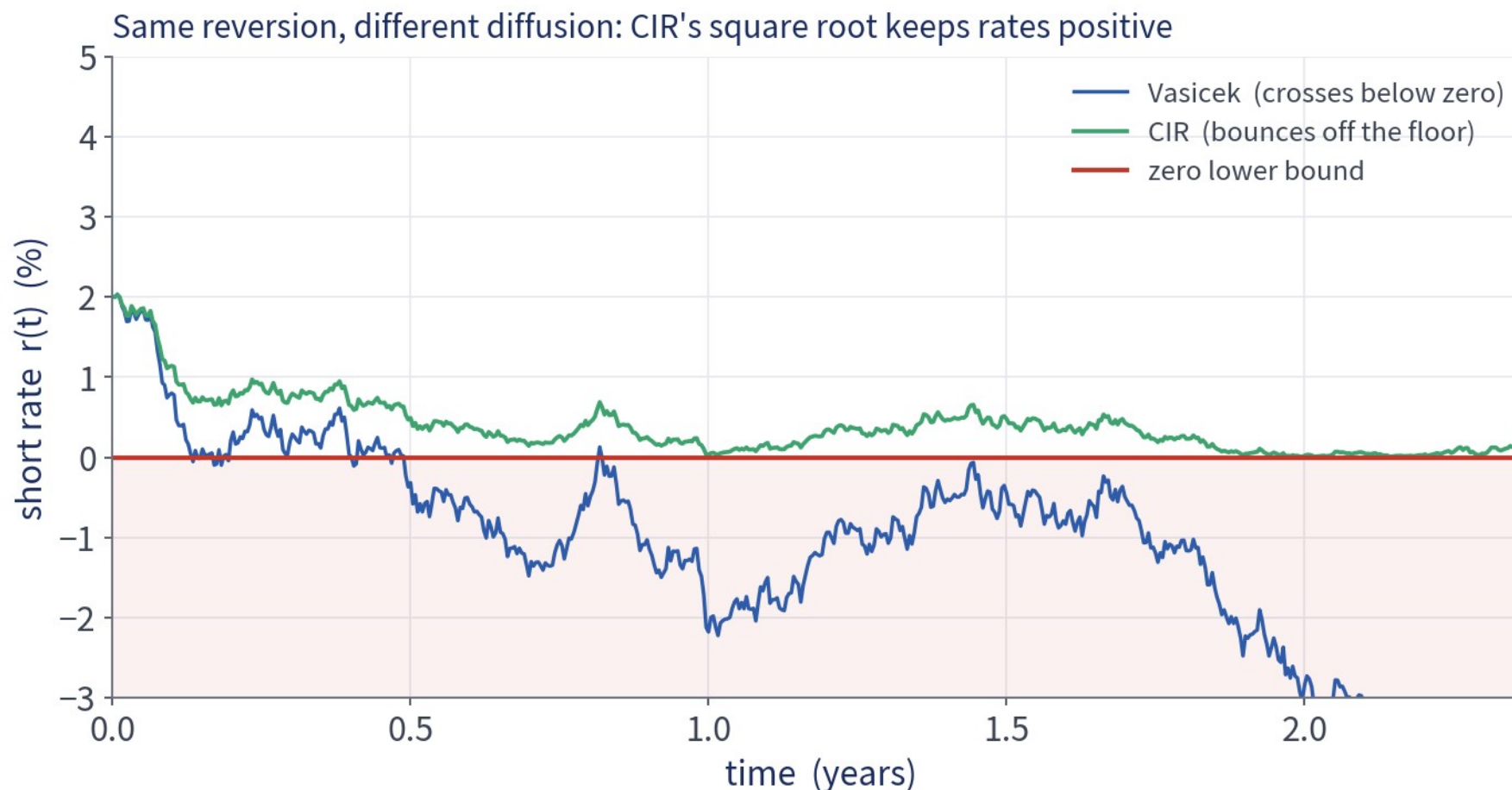
- If reversion and mean are strong enough relative to noise ...
- ... the rate is guaranteed to stay strictly positive.
- Formally, twice kappa times theta at least sigma squared.

Reading it

- Strong pull and a high mean keep the rate up.
- Large volatility fights against positivity.
- Meet the condition and zero is never reached.

CIR Bounces Off the Floor

From the same low start, Vasicek (blue) crosses below zero while CIR (green) is repelled by its vanishing volatility. Same reversion, different diffusion — and a very different result at the boundary.



CIR vs Vasicek at a Glance

Same reversion, different diffusion

Feature	Vasicek	CIR
Drift	$\kappa(\theta - r)$ — reverting	same — reverting
Diffusion	σ (constant)	$\sigma \sqrt{r}$ (level-dependent)
Negative rates	possible	ruled out (Feller)
Distribution	Gaussian	non-central chi-squared
Bond prices	closed-form	closed-form, harder

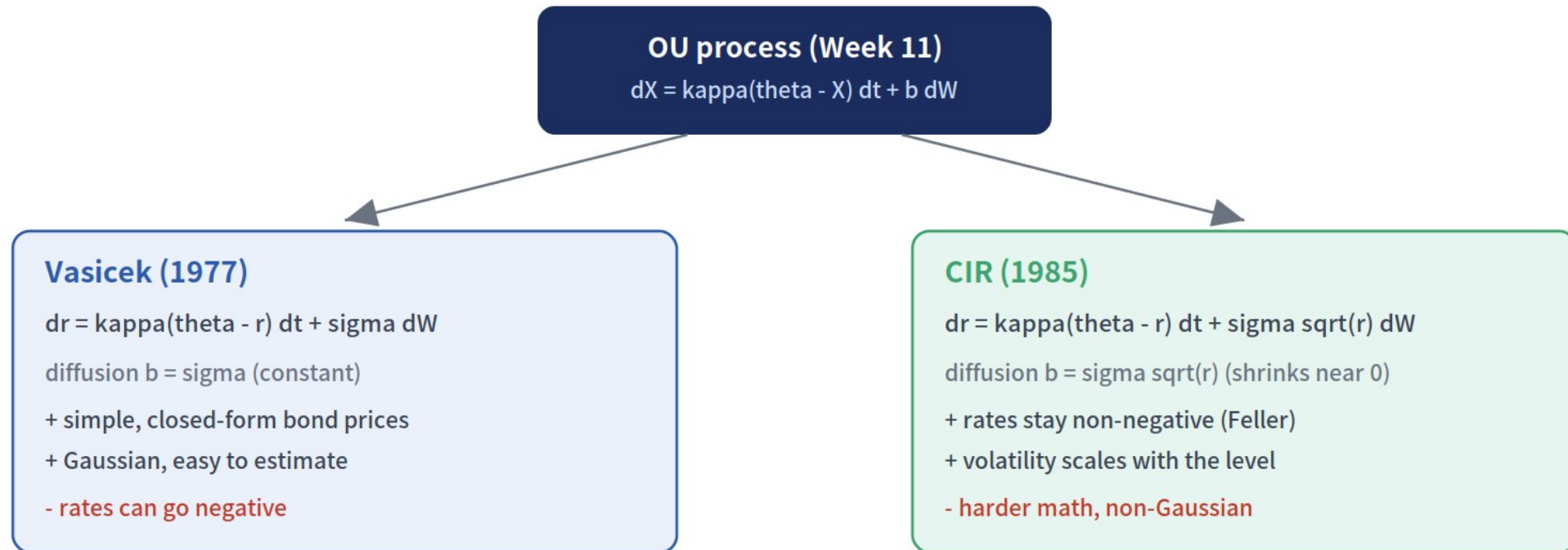
Both share the mean-reverting drift; the diffusion is the single design choice that sets them apart.

Short-Rate Models Are OU Cousins

Both descend from the OU process with the same mean-reverting drift. Vasicek keeps a constant diffusion; CIR uses a square-root diffusion that vanishes at zero. The drift is shared; the noise is the choice.

Short-Rate Models Are OU Cousins

Same mean-reverting drift as last week - the models differ only in their diffusion.



Same drift, different diffusion: the choice of $b(r)$ is what separates the two classic short-rate models.

When to Use Which

Trade-offs between the two

Reach for Vasicek

- You want the simplest, most tractable model.
- Gaussian rates and easy estimation are fine.
- The zero bound isn't a practical concern here.

Reach for CIR

- You need rates to stay strictly positive.
- You want volatility to scale with the level.
- You can accept somewhat harder mathematics.

Both Are OU Cousins

One idea, two diffusions

What they share

- The same mean-reverting drift, κ times the gap.
- A long-run level θ and a reversion speed κ .
- The whole OU intuition from Week 11.

What differs

- Only the diffusion term: constant vs square-root.
- That one choice decides the zero-bound behavior.
- Everything else is the OU machinery you know.

What Parts 2 and 3 Add

From the short rate to prices and code

Part 2 — the term structure

- How one short rate implies a whole yield curve.
- Discounting the random future rate into bond prices.
- Normal, inverted, and humped curve shapes.

Part 3 — computation

- Discretizing these SDEs (Euler and Milstein).
- Calibrating the models to market data.
- Pricing a bond end-to-end by simulation.

Simulate Vasicek and CIR

Direct Claude Code to see the zero floor

GOAL Simulate both short-rate models and watch how CIR's square-root diffusion keeps rates positive.

[INPUT] Prompt · Claude Code

Write and run `short_rates.py` that:

1. simulates Vasicek $dr = \kappa(\theta - r)dt + \sigma dW$ and CIR $dr = \kappa(\theta - r)dt + \sigma \sqrt{r} dW$ (Euler steps);
2. uses a low θ and high σ so Vasicek can go negative; plots several paths of each;
3. counts how often each model goes below zero;
4. checks the CIR Feller condition $2\kappa\theta \geq \sigma^2$ and reports whether it holds.

Show Vasicek dipping below zero while CIR does not.

How to work with it

- Both share the same reverting drift.
- Only the diffusion term differs.
- CIR's vol vanishes as r approaches zero.
- Feller holding means rates stay positive.

Deliverable: `short_rates.py` + Vasicek and CIR paths, a zero-crossing count, and a Feller check.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 What process is the Vasicek model, applied to the short rate?

The OU process

Q2 Why can Vasicek rates go negative?

Constant diffusion

Q3 How does CIR keep rates non-negative?

Square-root diffusion

All three come from the OU link, the constant diffusion, and the CIR square-root term.

Three Misconceptions to Avoid

Where rate-model intuition slips

Common myths

- "You can model rates like a stock with GBM."
- "Vasicek and CIR have different mean reversion."
- "CIR can never, under any parameters, hit zero."

The reality

- Rates revert; GBM trends — use a mean-reverting SDE.
- Their DRIFT is identical; only the diffusion differs.
- CIR stays positive only if the Feller condition holds.

Three Things to Carry Forward

Part 1 in one screen

1

Rates revert — so model the short rate

Interest rates are pulled toward a long-run level rather than trending. The short-rate approach models one instantaneous rate, and everything else follows from its dynamics.

2

Vasicek is OU for the rate

The same mean-reverting SDE from Week 11, now describing the short rate. Tractable and Gaussian — but its constant diffusion lets rates fall below zero.

3

CIR fixes the floor with a square root

Keeping the reverting drift but scaling volatility by the root of the rate, CIR makes shocks vanish near zero. Under the Feller condition, rates stay strictly positive.

Next — Part 2: from the short rate to bond prices and the shapes of the yield curve.